CLASS 4

Nuclear & HEP Astrophysics

A First Exposure for Particle Physicists

Duration · 3 months · 13 weeks
Weekly Time · ~5–10 hrs (running parallel with one class)
Sequence · 2026–2027 Self-Study Plan (RHI → HEP)

Deliberately broad tour of the astrophysics most connected to nuclear and HEP. Starts from zero astronomy background.

Course Overview

Labeled “first exposure” on purpose. You have never taken an astronomy class; this course starts there and builds a working map of the nuclear and HEP astrophysics landscape. The “Astronomy Primer” section is your orientation — read it before Week 1 proper.

The goal is literacy and motivation. You should be able to: (a) follow an astro colloquium, (b) understand how nuclear/HEP techniques are used in astrophysics, and (c) identify one or two topics you might follow deeper later.

For an RHI physicist, there are obvious bridges: lattice QCD EOS for neutron-star interiors, heavy-ion-style simulations of NS mergers, hypernuclear physics. Note these as you go.

Learning Goals

- Use astronomy's scales, units, and taxonomy fluently.
- Describe stellar structure and the major reaction networks (pp, CNO, s-, r-, rp-).
- Explain compact objects at the M-R and EOS-constraint level.
- Follow a cosmic-ray or astro-ν paper.
- Describe the standard cosmological model at the particle-content level.
- Identify specific RHI/HEP $\leftrightarrow$ astrophysics bridges.

Prerequisites

Deliberately light. The primer covers the astronomy vocabulary.

- Basic stat mech and thermodynamics.
- Nuclear physics concepts (binding energy, cross sections, decay).
- Rough GR picture (Schwarzschild, horizons).
- **No prior astronomy required** — the primer handles it.

Astronomy Primer

If you have never taken an astronomy class, start here. This section front-loads the vocabulary, scales, and observational concepts that the rest of the course assumes. It is not a replacement for the Week 1–2 reading in Carroll & Ostlie, but it is enough to make that reading make sense.

1. Scales and units

Astronomy uses units tailored to the scales it studies. You will see these constantly — internalize them:

- **Astronomical Unit (AU)** = mean Earth-Sun distance ≈ 1.496×10^{11} m. Used for solar-system scales.
- **Parsec (pc)** = distance at which 1 AU subtends 1 arcsec ≈ 3.086×10^{16} m ≈ 3.26 light-years. The nearest star is 1.3 pc away. The Milky Way disc is ~30 kpc across.
- **Megaparsec (Mpc)** = 10^6 pc. Galaxies are typically tens of kpc; distances between galaxies are Mpc; the observable universe is about 14,000 Mpc.
- **Solar mass ($M_\odot$)** = 1.989×10^{30} kg. Stars range from ~0.08 to ~100 $M_\odot$.
- **Solar luminosity ($L_\odot$)** = 3.83×10^{26} W. Used as the unit for stellar brightness.
- **Magnitudes.** Logarithmic scale, smaller = brighter. 5 magnitudes = factor of 100 in brightness. Absolute magnitude is measured at 10 pc. Annoying but entrenched; you will learn to translate on the fly.

2. The electromagnetic spectrum

Almost everything in astronomy is inferred from electromagnetic radiation. Different wavelengths tell you about different physics. You need to know the rough ranges:

- **Radio** (>1 mm): interstellar gas, pulsars, CMB. Atmospherically transparent.
- **Microwave / mm / sub-mm**: cold dust, CMB, molecular lines.
- **Infrared**: cool objects (~10–1000 K), dusty regions, exoplanets. Mostly blocked by atmospheric water; space telescopes preferred (JWST).
- **Optical** (~400–700 nm): main-sequence stars, historically the default.
- **Ultraviolet**: hot stars, supernovae, galactic nuclei. Blocked by ozone.
- **X-ray**: accretion onto compact objects, hot gas ($T > 10^6$ K). Space only (Chandra, XMM-Newton).
- **Gamma-ray**: cosmic rays, pulsars, GRBs, decay lines. Space (Fermi-LAT) or air-Cherenkov arrays (HESS, CTA).
- **Gravitational waves** (not EM): mergers of compact objects. LIGO/Virgo/KAGRA.
- **Neutrinos** (not EM): solar, supernova, astrophysical. IceCube, SNO+.

3. Stellar classification and the HR diagram

Stars are classified by spectral type (O B A F G K M, hot to cool; famously remembered as “Oh Be A Fine Girl/Guy, Kiss Me”) and luminosity class (I supergiant, II bright giant, III giant, IV subgiant, V main-sequence, etc.). The Sun is a G2V. The Hertzsprung-Russell (HR) diagram plots luminosity vs temperature (or spectral type); most stars lie on the *main sequence*, where they fuse hydrogen into helium in their cores.

The three key regions of the HR diagram

Main sequence: the diagonal band. Luminosity scales as ~$M^{3.5}$. Lifetime scales as M / L ~ $M^{-2.5}$: massive stars live fast and die young.

Giants / Supergiants: top right. Evolved stars that have exhausted core hydrogen and are burning heavier fuels (He, C, O) in shells and cores. Large, cool, luminous.

White dwarfs: bottom left. Degenerate stellar remnants of low/intermediate-mass stars. Hot, tiny, dim.

4. The distance ladder

A critical concept: no single technique works at all distances. We build up a *ladder* where each rung calibrates the next:

- **Parallax** (< ~10 kpc): geometric. Measure the apparent shift of a star against background as Earth orbits. Gaia has revolutionized this.
- **Standard candles** (main sequence fitting, Cepheid variables, RR Lyrae): objects whose intrinsic luminosity is known, so apparent brightness gives distance.
- **Type Ia supernovae** (up to ~1 Gpc): very bright standardizable candles. Their Hubble diagram gave us dark energy.
- **Redshift + Hubble's law** (cosmological distances): $cz \approx H_0 d$ for small z, with H_0 the Hubble constant (~67–73 km/s/Mpc; the tension between measurements is a current major issue).

5. What types of objects are out there?

- **Stars** at every stage of life: main sequence, red giants, supergiants, variable stars (Cepheids, RR Lyrae, Mira).
- **Stellar remnants**: white dwarfs (~0.6 $M_\odot$, Earth-sized), neutron stars (~1.4 $M_\odot$, 10 km radius), black holes (stellar mass a few to ~100 $M_\odot$, and supermassive $10^{6–10}$ $M_\odot$ at galactic centers).
- **Binaries**: the majority of stars. Source of a huge fraction of the interesting physics: X-ray binaries, cataclysmic variables, gravitational-wave events.
- **Interstellar medium (ISM)**: gas and dust between the stars, in many phases (cold molecular, warm neutral, hot ionized).
- **Star-forming regions**: dense molecular clouds collapsing into new stars.
- **Planetary systems**: thousands confirmed via transits (Kepler, TESS) and radial velocities.
- **Galaxies**: ~10^{11} stars each, classified by Hubble's morphological sequence (ellipticals, spirals, irregulars). Most have a supermassive black hole at their center.
- **Active galactic nuclei (AGN)**: accreting supermassive black holes that outshine their host galaxies. Quasars are the most luminous.
- **Galaxy clusters**: largest bound structures. Filled with hot X-ray-emitting intracluster gas.
- **Cosmic microwave background (CMB)**: relic radiation from when the universe became transparent, ~380,000 years after the Big Bang. Nearly uniform at ~2.725 K with ~10^{-5} anisotropies.

6. Basic cosmology vocabulary

- **Big Bang**: the universe began ~13.8 Gyr ago as a hot, dense state and has been expanding and cooling since.
- **Redshift (z)**: $(\lambda_{obs} - \lambda_{emit})/\lambda_{emit}$. For cosmological distances, measures the expansion of space since the light was emitted. $z = 1$ means the universe was half its current size; the CMB is at $z \approx 1100$.
- **Dark matter**: non-luminous matter making up ~27% of the universe's mass-energy. Evidence: galaxy rotation curves, CMB, gravitational lensing. Nature unknown.
- **Dark energy**: ~68% of mass-energy. Drives accelerated expansion. Cosmological constant Λ is the simplest model; quintessence is an alternative.
- **Big Bang nucleosynthesis (BBN)**: in the first few minutes, light elements (H, He, Li) were produced. Predictions match observations; a pillar of standard cosmology.
- **Inflation**: a conjectured period of very rapid expansion in the first 10^{-32} s that explains flatness, horizon, and monopole problems and seeds structure.

7. How you'll connect to this

Two places where your RHI / HEP training becomes directly useful in astrophysics, which you should watch for as the course progresses:

- **Dense-matter equations of state.** Neutron star interiors reach densities where QCD matters. Lattice QCD, heavy-ion collisions, and neutron-star observations (masses, radii, merger GW signals) constrain the same equation of state. This is an active interface.

- **Astrophysical neutrinos and cosmic rays.** Detection is done with particle-physics-style detectors (IceCube uses photomultipliers in ice; Auger uses water-Cherenkov surface detectors). The analysis culture is closer to HEP than to classical astronomy.
- **Multimessenger astronomy.** Combined EM + GW + neutrino observations of single events (e.g., GW170817) are the new mode. Particle physicists fit naturally into these collaborations.

Textbooks & Resources

Primary Texts

- **Carroll & Ostlie**, *An Introduction to Modern Astrophysics* — this is where your reading begins; spine of Weeks 1–2.
- **Iliadis**, *Nuclear Physics of Stars* — spine of Weeks 3–7.
- **Perkins**, *Particle Astrophysics* — spine of Weeks 8–12.

Supplementary / Reference

- **Rolfs & Rodney**, *Cauldrons in the Cosmos*.
- **Gaisser, Engel & Resconi**, *Cosmic Rays and Particle Physics*.
- **Shapiro & Teukolsky**, *Black Holes, White Dwarfs, and Neutron Stars*.
- **Kolb & Turner**, *The Early Universe*.
- **Dodelson & Schmidt**, *Modern Cosmology*.

Online Lectures & Notes

- JINA-CEE webinars (joint nuclear astrophysics).
- NASA ADS — start searching papers here.
- NNDC databases (nndc.bnl.gov) — cross sections.
- IceCube, Pierre Auger, LIGO/Virgo, KM3NeT public pages.
- Living Reviews in Relativity & Solar Physics.
- Annual Reviews of Nuclear and Particle Science.

Key Papers & Reviews

- B^2FH, *Synthesis of the Elements in Stars* (1957).
- Bahcall et al., solar neutrino reviews.
- IceCube, diffuse astrophysical neutrino papers.
- LIGO+Virgo, GW170817 multimessenger (2017).
- Kasen et al., kilonova nucleosynthesis.
- Planck 2018 cosmology parameter paper.
- XENONnT, LZ recent dark-matter limit papers.

13-Week Schedule

First two weeks are deliberately gentle — they are your astronomy introduction. Pace picks up from Week 3.

Week	Topic	Readings & Activities	Deliverable
1	Intro to astronomy; scales, units, HR diagram	Primer; C&O Chs. 1-3, 8	HR diagram from scratch
2	Stellar structure essentials	C&O Chs. 10-11; Iliadis Ch. 1	Derive Lane-Emden polytropes
3	Stellar evolution overview	C&O Chs. 13-15	HR diagram annotated
4	Nuclear reactions in stars I	Iliadis Chs. 2-3; Rolfs-Rodney	pp-chain rate at T_c
5	Nuclear reactions in stars II	Iliadis Chs. 4-5	CNO cycle flow diagram
6	s-process, r-process	Iliadis Ch. 5; review papers	r-process site notes
7	Supernovae; compact object formation	C&O Ch. 15; Bethe reviews	ν-driven mechanism summary
8	Compact objects: WDs, NSs, BHs	Shapiro-Teukolsky Chs. 2-3, 8-9	Chandrasekhar derivation
9	Mergers & multimessenger	S-T Ch. 12; GW170817 paper	Kilonova / r-process link
10	Cosmic rays	Perkins Chs. 6-7; Gaisser excerpts	CR spectrum annotated
11	Astrophysical neutrinos; dark matter	Perkins Chs. 8-9; IceCube papers	IceCube sensitivity sketch
12	Cosmology essentials	Dodelson-Schmidt Chs. 1-2; Planck 2018	Freeze-out Boltzmann eq
13	Multimessenger synthesis; capstone	GW170817 companion papers	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other. If time is tight, cut problem sets before cutting reading; if loose, add paper depth.

Problem Sets & Exercises

Lighter than the core classes. One or two per week.

- Iliadis problems in Chs. 2–5 — those with astro numerics.
- Compute nuclear-burning lifetime for a 1 $M_\odot$ main-sequence star.
- Derive the Chandrasekhar mass; discuss composition independence.
- Estimate ν flux on Earth from a Galactic core-collapse SN vs Super-K.
- Plot the CR spectrum on log-log; annotate knee, ankle, GZK.
- Derive WIMP thermal relic abundance (Kolb-Turner).

Capstone Project

~6 pages, focused. Pick a nuclear/HEP × astro bridge and write the review you wish existed.

- Pick a topic: kilonova r-process; NS EOS from lattice QCD; sterile neutrinos in cosmology; axion DM searches; MM constraints on NS merger remnants.
- Motivate why it matters for particle physics.
- Describe relevant astro observations and theoretical machinery.
- Identify 1–2 open questions and the measurements that address them.
- Conclude with: “How would my RHI/HEP skills contribute?”

Milestones & Self-Assessment

At each checkpoint you should do the listed tasks without looking them up. If you can't, go back.

- **End of Week 1:** read an astro paper abstract and understand the scales and object types without lookup.
- **End of Week 5:** sketch the H and He burning networks; explain why CNO dominates in massive stars.
- **End of Week 9:** explain the TOV equation, NS maximum mass, why mergers are the r-process site.
- **End of Week 13:** follow an astro colloquium; list concrete research directions you could imagine working on.

End of class plan. Iterate after the first two weeks.